Multimodal Evidence

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Multimodal Evidence before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Speech and Audio Systems

Without opening the old lesson, explain one key idea from Speech and Audio Systems. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Image Generation Models

Without opening the old lesson, explain one key idea from Image Generation Models. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Iterative Prompting

Without opening the old lesson, explain one key idea from Iterative Prompting. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Multimodal Evidence trace and record the first step you can explain without help.

Use W2: Trace the worked case for Speech pipelines and multimodal **evidence**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Multimodal Evidence explanation against the supplied facts before discussing it.

Use the worked case for Speech pipelines and multimodal **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

> A description reports two visible plants and a third opaque pot whose contents cannot be seen. A caption says “three plants”. What is supported?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Multimodal Evidence target to a trace and a justified conclusion.

A caption says four seedlings; the image shows three and an opaque box. Is a fourth seedling disproved? Write an **evidence**-aware response.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Multimodal Evidence case and show the deciding evidence.

An image shows three plants; caption claims a fourth outside the frame.

What can the image establish?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

A chart image shows 12 plants; its caption says 21. What should a multimodal assistant report before drawing a conclusion?

My independent answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Multimodal Evidence adds to Verified Study Buddy and keep the evidence.

Accept diagrams plus questions only when the source and topic align with the learner's task.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A caption says four seedlings; the image shows three and an opaque box. Is a fourth seedling disproved? Write an **evidence**-aware response.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Multimodal Evidence facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Multimodal Evidence", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace the worked case

> Trace the worked case. Apply the supplied Multimodal Evidence facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Multimodal Evidence", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Audit the limit

> Audit the limit. Apply the supplied Multimodal Evidence facts and keep your reasoning.

A report omits this qualification: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Multimodal Evidence facts and keep your reasoning.

Transcript says room E; a current written approval says room F. The task requires current approved room. Written approval is declared authoritative; trace the response and the **transcription** limit.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Multimodal Evidence facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Multimodal Evidence", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Multimodal Evidence

> A fictional Study Buddy audits a desk description and a separate caption. Explain the labelled step before checking the changed condition.

A fictional Study Buddy audits a desk description and a separate caption.

Description: one open transparent box visibly contains three pencils; one closed opaque box hides its contents.

Caption: “There are five pencils on this desk.” No other **evidence** reveals the closed box.

AI ACADEMY / AI-3/13

Multimodal Evidence

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Visible evidence

Three pencils are explicitly visible in the open box.

2. Hidden region

The closed box's pencil count is unknown.

3. Caption claim

Five total requires two additional pencils if these are the only relevant locations.

4. Supported response

At least three are supported; total five is not established by the description.

CHANGE ONE THING

Open the closed box and a verified observation shows two pencils inside, with no other desk locations.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why were both “exactly three” and “exactly five” unsupported before opening the box?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Label Study Buddy claims by their visual, audio or text **evidence** and retain uncertainty when modalities omit information.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Label Study Buddy claims by their visual, audio or text **evidence** and retain uncertainty when modalities omit information.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Study Buddy audits a desk description and a separate caption.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? At least three are supported; total five is not established by the description.

1. Caption claim

2. Supported response

3. Hidden region

4. Visible **evidence**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Supported response At least three are supported; total five is not established by the description.

A caption is another claim, not a way of seeing through an opaque object. Keep observable information separate from inferences.

The visible three give a lower bound under the stated desk scope; there is no supplied maximum for the closed box. Do not replace unknown contents with zero merely because none are visible.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Open the closed box and a verified **observation** shows two pencils inside, with no other desk locations. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. The total is now supported as five.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The total is now supported as five. 3 visible initially plus 2 newly observed gives 5 under the complete-location assumption.

Repair a missing building block

Review Speech and Audio Systems (ai-3/12). Goal: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**.

Short practice: Which stage was initially wrong, and which stage was correctly operating on a wrong **input**?. Return to this week and explain what you changed.

Review Image Generation Models (ai-3/10). Goal: Explain the **mechanism** and terminology of image generation and provenance.

Short practice: What may these two grids explain, and what can they not establish about a real generator?. Return to this week and explain what you changed.

Review Iterative Prompting (ai-3/7). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: What **evidence** supports improvement here, and what remains unproven?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

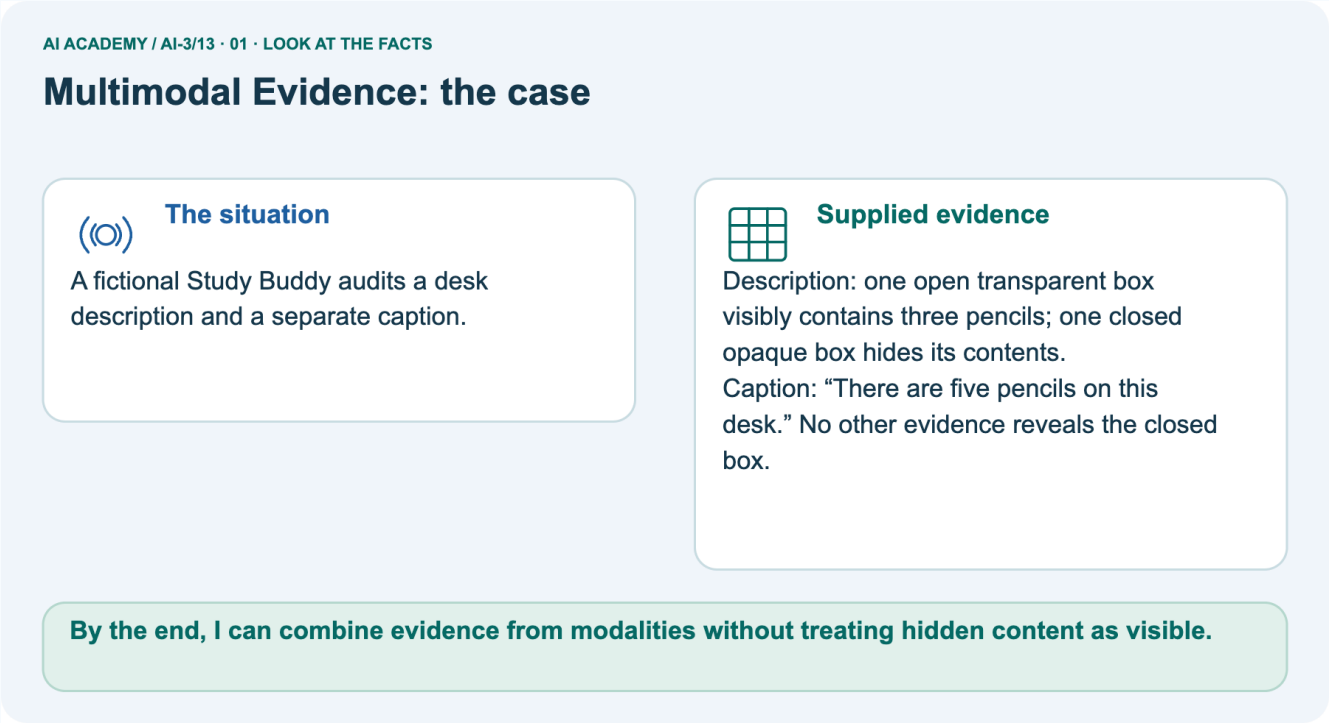
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Multimodal Evidence picture from its labels, then record a relevant limit.

01. Multimodal Evidence: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional Study Buddy audits a desk description and a separate caption. Supplied facts: Description: one open transparent box visibly contains three pencils; one closed opaque box hides its contents.

Caption: “There are five pencils on this desk.” No other **evidence** reveals the closed box.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



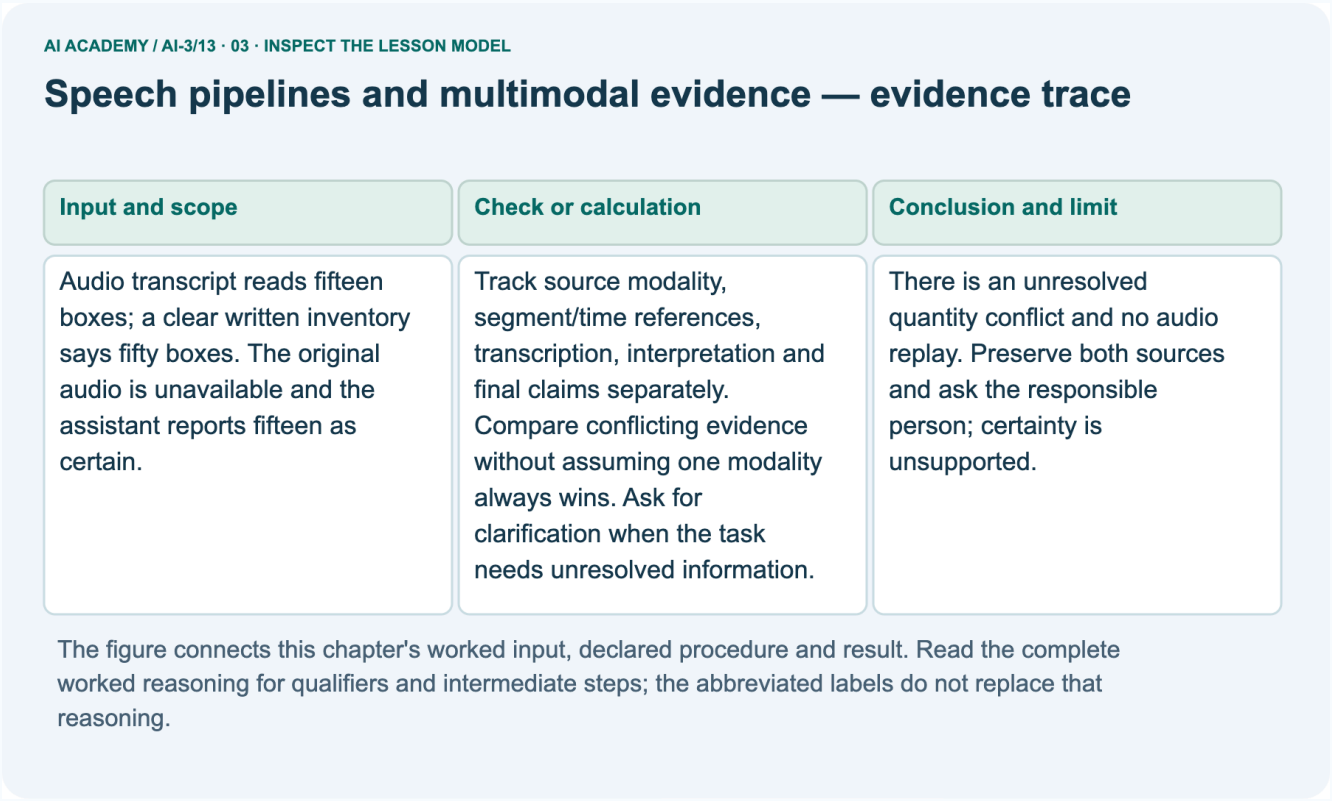
Follow the labelled picture with the matching explanation. Open the vector diagram

Visible **evidence**: Three pencils are explicitly visible in the open box. → Hidden region: The closed box's pencil count is unknown. → Caption claim: Five total requires two additional pencils if these are the only relevant locations. → Supported response: At least three are supported; total five is not established by the description.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Speech pipelines and multimodal evidence — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Speech pipelines and multimodal **evidence** — **evidence** trace. Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes.

The original audio is unavailable and the assistant reports fifteen as certain. | Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately.

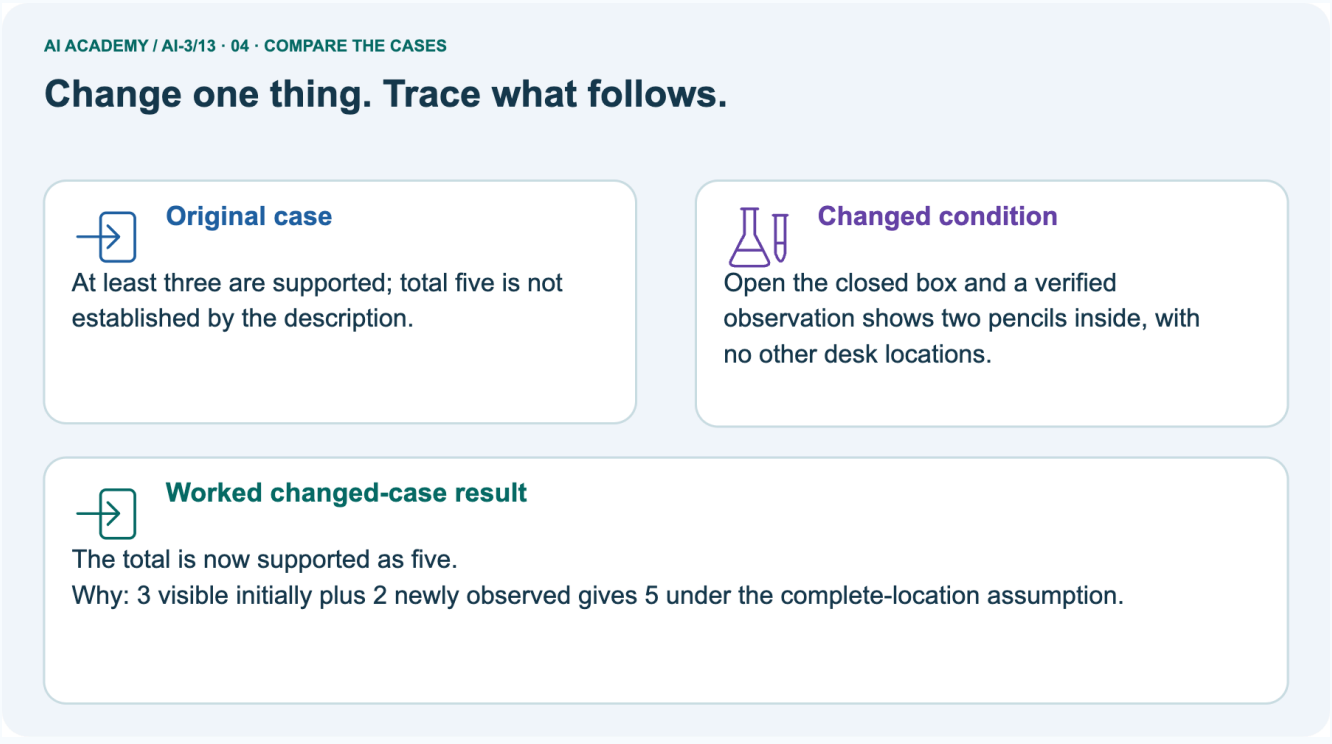
Compare **conflict**ing **evidence** without assuming one **modality** always wins. Ask for clarification when the task needs unresolved information. | There is an unresolved quantity **conflict** and no audio replay.

Preserve both sources and ask the responsible person; certainty is unsupported.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: At least three are supported; total five is not established by the description. Changed condition: Open the closed box and a verified **observation** shows two pencils inside, with no other desk locations.

Expected result: The total is now supported as five. Reason: 3 visible initially plus 2 newly observed gives 5 under the complete-location assumption.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit

AI ACADEMY / AI-3/13 · 05 · EXPLAIN WITH CARE

Connect the explanation to its limit

What the case explains

A caption is another claim, not a way of seeing through an opaque object. Keep observable information separate from inferences. The visible three give a lower bound under the stated desk scope; there is no supplied maximum for the closed box. Do not replace unknown contents with zero merely because none are visible.

What it does not establish

The conclusion depends on the supplied description's accuracy; no actual image inspection or exhaustive room inventory was performed.

Follow the labelled picture with the matching explanation. Open the vector diagram

A caption is another claim, not a way of seeing through an opaque object. Keep observable information separate from inferences.

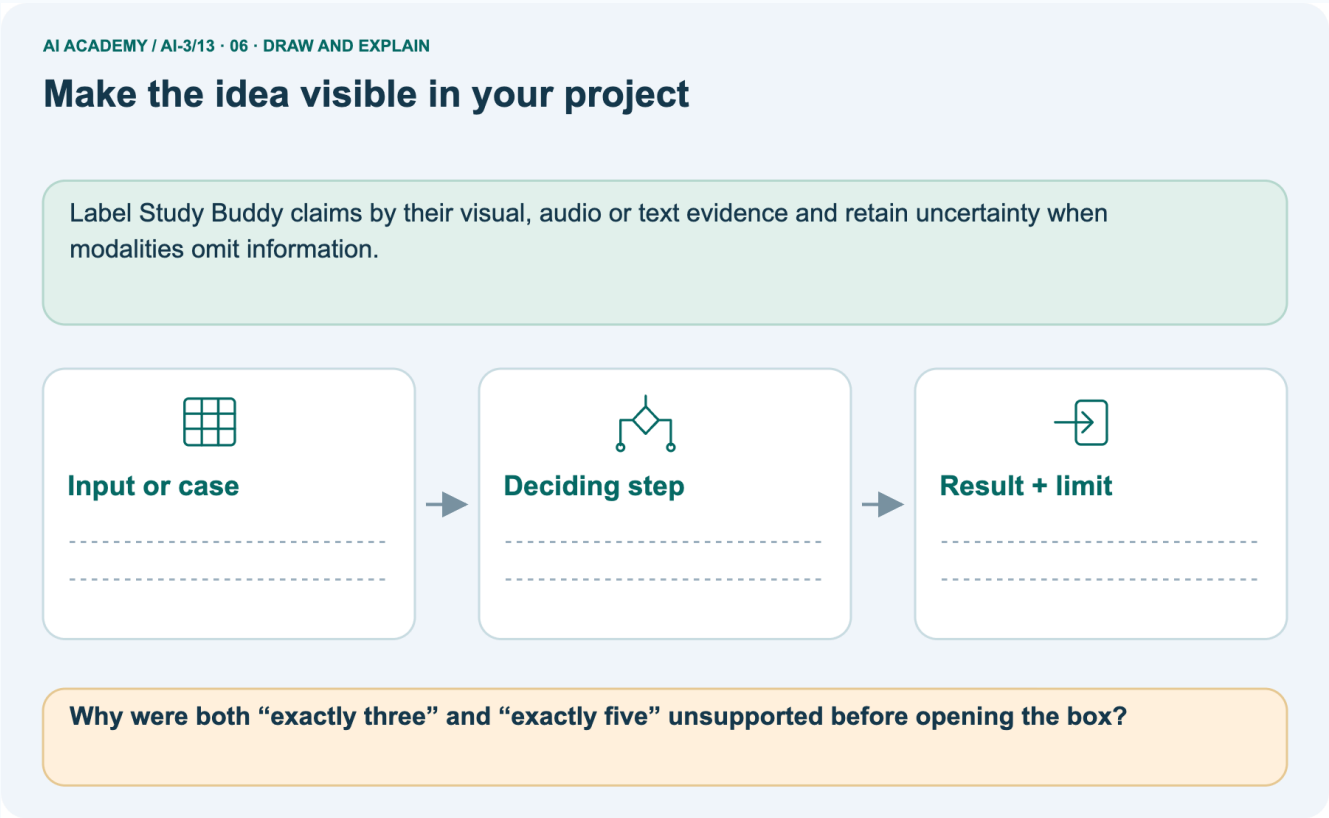
The visible three give a lower bound under the stated desk scope; there is no supplied maximum for the closed box. Do not replace unknown contents with zero merely because none are visible.

Limit: The conclusion depends on the supplied description’s accuracy; no actual image inspection or exhaustive room inventory was performed.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Label Study Buddy claims by their visual, audio or text **evidence** and retain uncertainty when modalities omit information. Draw your **input**, deciding step, result and limit.

Supported question: Why were both “exactly three” and “exactly five” unsupported before opening the box?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.